

Loan Compliance System

Header/toolbar alignment fix — change log and setup instructions

Covers app.py and requirements.txt as revised in this conversation.

1. What was broken

The header (logo/title, the “OPA Rego + AI Agent” pill, the four live stat chips, and the Import/Reset/Export buttons) and the Review Workdesk toolbar (search box + Rules/Review/Rows filters) both used Streamlit's

```
st.columns([3.4, 1.5, 1, 0.9, 1, 1.05, 1, 1, 1.15])
```

Fixed-ratio columns like this **shrink but never wrap**. On any browser window narrower than one specific wide viewport, this caused two visible problems:

- The “OPA Rego + AI Agent” pill (styled with white-space: nowrap) overflowed its shrinking column and rendered on top of the “Total Audited” stat next to it.
- The Import CSV / Reset 100k / Export Audit CSV buttons (use_container_width=True inside an increasingly narrow column) had their labels wrap letter-by-letter.

Neither is a rendering glitch that fixes itself - it happens on any screen or window narrower than the one the columns ratios were tuned for.

2. What changed

The fix replaces fixed-ratio columns with Streamlit's native horizontal flex container - `st.container(horizontal=True, ...)` - which, per Streamlit's own documentation, wraps items onto a new line once they no longer fit, instead of shrinking them below their natural size.

Where	Before	After
<code>render_header()</code>	Fixed-ratio <code>st.columns([3.4, 1.5, 1, ...])</code> - 9 columns	Single flat <code>st.container(horizontal=True, ...)</code> - no nested sub-containers
<code>_render_workdesk_toolbar()</code>	Fixed-ratio <code>st.columns([3, 1.3, 1.3, 0.9])</code>	<code>st.container(horizontal=True, ...)</code> with search box set to stretch and the 3 selects at fixed pixel widths
Header buttons	<code>use_container_width=True</code> (forced to fill a shrinking column)	Default content width - buttons size to their own label
<code>requirements.txt</code>	<code>streamlit>=1.38</code>	<code>streamlit>=1.48</code> (minimum that supports horizontal flex containers)
Page width	max-width: 1680px	max-width: 1800px (content needs ~1705px to sit on one line)

Key implementation detail: the header buttons no longer use `use_container_width=True`. That flag was what forced each button to fill its (shrinking) column, which is what caused the letter-by-letter wrapping. Left at their natural content width, buttons simply move to a new line as a group when space runs out.

3. Bugs found and fixed while testing against your exact environment

The app was re-tested against **Streamlit 1.50.0** specifically (the newest version that installs on Python 3.9, matching your setup) rather than only the newer version used during initial development. That surfaced three additional issues, all now fixed in the delivered files:

Symptom	Cause	Fix
StreamlitInvalidColumnGapError: gap must be small/medium/large/none	gap=12 (an integer) was passed to st.container	Changed to gap="small" / gap="medium" - Streamlit 1.50.0 only accepts these string literals, not a pixel integer
StreamlitInvalidWidthError: width must be an integer or 'stretch'	A nested container used width="content"	Removed - "content" is only valid on leaf elements (buttons, popovers), not on st.container itself in 1.50.0
Buttons forced onto their own line even with room to spare	Buttons were nested inside their own st.container(), which defaults to width="stretch" (100% of parent)	Flattened - removed the nested container so buttons sit as direct, content-sized children of the outer row

Note: nested `st.container()` calls default their own width to “stretch” (100% of the parent). If you add further nested containers to this header in the future, keep that in mind - it is the one gotcha that is easy to reintroduce by accident.

4. How to run the app

```
pip install -r requirements.txt
streamlit run app.py
```

Minimum supported version is now `streamlit>=1.48` (bumped from 1.38), since the fix depends on horizontal flex containers, added in Streamlit 1.48. This is still compatible with Python 3.9, where 1.50.0 is the newest version pip can install.

5. Verification performed

- Ran the app under Streamlit 1.50.0 (installed in an isolated environment to mirror the target setup) and confirmed no errors on load or on any of the four tabs.
- Screenshoted the header and toolbar at 850, 1000, 1300, 1700, 1900, and 2200px viewport widths - confirmed items wrap onto additional lines cleanly with no overlapping text at any width.
- Confirmed the header sits on a single line at typical wide-desktop widths ($\geq \sim 1800$ px page width) and degrades to two or three lines gracefully below that, rather than overlapping.
- Clicked through all four tabs (Review Workdesk, Compliance & Portfolio Report, OPA Rego Playground, Asset Value Predictor) to confirm nothing else regressed.

6. If you see another error like this in future

Streamlit's horizontal-container layout API is relatively new (added in version 1.48), and some of its parameter constraints - which literal strings are valid for `gap`, whether `width` accepts "content" - differ slightly between minor versions. If you upgrade Streamlit and see a similar `StreamlitInvalidColumnGapError` or `StreamlitInvalidWidthError`, share the traceback and it can be adjusted the same way again.